Q1 - 24 January - Shift 1

Let PQR be a triangle. The points A, B and C are on the sides QR, RP and PQ respectively such that $\frac{QA}{AR}=\frac{RB}{BP}=\frac{PC}{CQ}=\frac{1}{2}$. Then $\frac{\text{Area}(\Delta PQR)}{\text{Area}(\Delta ABC)}$ is equal to

(1) 4 (2) 3

(3) 2 (4) $\frac{5}{2}$

Space for your notes:

Q2 - 24 January - Shift 2

The equations of the sides AB, BC and CA of a triangle ABC are: $2x + y = 0$, $x + py = 21a$, $(a \neq 0)$ and $x - y = 3$ respectively. Let P $(2, a)$ be the centroid of ΔABC. Then $(BC)^2$ is equal to

Space for your notes:

Q3 - 25 January - Shift 2

The equations of two sides of a variable triangle are $x = 0$ and $y = 3$, and its third side is a tangent to the parabola $y^2 = 6x$. The locus of its circumcentre is :

(1) $4y^2-18y-3x-18=0$ (2) $4y^2+18y+3x+18=0$

(3) $4y^2-18y+3x+18=0$ (4) $4y^2-18y-3x+18=0$

Space for your notes:

Q4 - 25 January - Shift 2

A triangle is formed by X – axis, Y– axis and the line $3x + 4y = 60$. Then the number of points P(a, b)which lie strictly inside the triangle, where a is an integer and b is a multiple of a, is_____.

Space for your notes:

Q5 - 29 January - Shift 1

A light ray emits from the origin making an angle 30º with the positive x-axis. After getting reflected by the line $x + y = 1$, if this ray intersects x-axis at Q, then the abscissa of Q is

(1) $\frac{2}{(\sqrt{3}-1)}$ (2) $\frac{2}{3+\sqrt{3}}$

(3) $\frac{2}{3-\sqrt{3}}$ (4) $\frac{\sqrt{3}}{2(\sqrt{3}+1)}$

Space for your notes:

Q6 - 29 January - Shift 1

Let B and C be the two points on the line $y + x = 0$ such that B and C are symmetric with respect to the origin. Suppose A is a point on $y - 2x = 2$ such that ΔABC is an equilateral triangle. Then, the area of the ΔABC is

(1) $3\sqrt{3}$ (2) $2\sqrt{3}$

(3) $\frac{8}{\sqrt{3}}$ (4) $\frac{10}{\sqrt{3}}$

Space for your notes:

Q7 - 29 January - Shift 2

A triangle is formed by the tangents at the point (2, 2) on the curves $y^2 = 2x$ and $x^2 + y^2 = 4x$, and the line $x + y + 2 = 0$. If r is the radius of its circumcircle, then r^2 is equal to ______.

Space for your notes:

Q8 - 30 January - Shift 1

A straight line cuts off the intercepts $OA = a$ and $OB = b$ on the positive directions of x–axis and y–axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of y–axis and the area of ΔOAB is $\frac{98}{3}\sqrt{3}$, then $a^2 - b^2$ is equal to:

(1) $\frac{392}{3}$ (2) 196

(3) $\frac{196}{3}$ (4) 98

Space for your notes:

Q9 - 01 February - Shift 1

If the orthocentre of the triangle, whose vertices are (1, 2), (2, 3) and (3, 1) is (α, β), then the quadratic equation whose roots are $\alpha + 4\beta$ and $4\alpha + \beta$, is

(1) $x^2 - 19x + 90 = 0$
(2) $x^2 - 18x + 80 = 0$
(3) $x^2 - 22x + 120 = 0$
(4) $x^2 - 20x + 99 = 0$

Space for your notes:

Q10 - 01 February - Shift 1

The combined equation of the two lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ can be written as $(ax + by + c)(a'x + b'y + c') = 0$

The equation of the angle bisectors of the lines represented by the equation $2x^2 + xy - 3y^2 = 0$ is

(1) $3x^2 + 5xy + 2y^2 = 0$
(2) $x^2 - y^2 + 10xy = 0$
(3) $3x^2 + xy - 2y^2 = 0$
(4) $x^2 - y^2 - 10xy = 0$

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (2) **Q2** (122) **Q3** (3) **Q4** (31)

Q5 (2) **Q6** (3) **Q7** (10) **Q8** (1)

Q9 (4) **Q10** (4)

Q1 (2)

Let P is $\vec{0}$, Q is $\vec{q}$ and R is $\vec{r}$

A is $\frac{2\vec{q}+\vec{r}}{3}$, B is $\frac{2\vec{r}}{3}$ and C is $\frac{\vec{q}}{3}$

Area of ΔPQR is $= \frac{1}{2}\left|\vec{q}\times\vec{r}\right|$

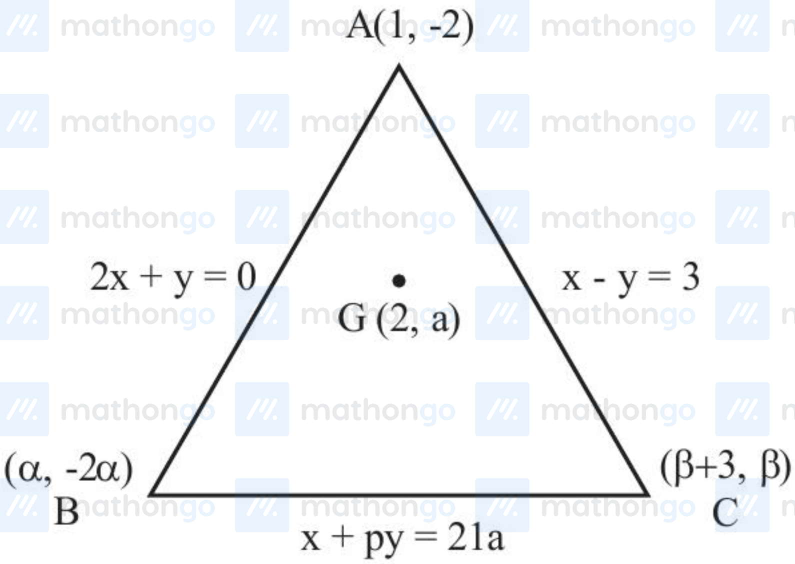
Area of ΔABC is $\frac{1}{2}\left|\overrightarrow{AB}\times\overrightarrow{AC}\right|$

$\overrightarrow{AB} = \frac{\vec{r}-2\vec{q}}{3}, \overrightarrow{AC} = \frac{-\vec{r}-\vec{q}}{3}$

Area of $\Delta ABC = \frac{1}{6}\left|\vec{q}\times\vec{r}\right|$

$$\frac{\text{Area}(\Delta PQR)}{\text{Area}(\Delta ABC)} = 3$$

Q2 (122)



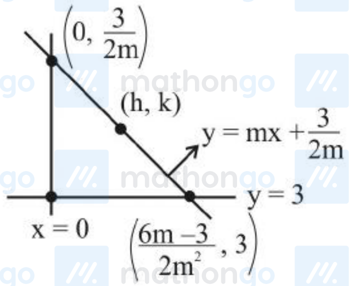
Assume $B(\alpha, -2\alpha)$ and $C(\beta + 3, \beta)$

$\frac{\alpha+\beta+3+1}{3} = 2$ also $\frac{-2\alpha-2+\beta}{3} = a$

Q3 (3)

$y^2 = 6x \quad \& \; y^2 = 4ax$

$\Rightarrow 4a = 6 \Rightarrow a = \dfrac{3}{2}$



$y = mx + \dfrac{3}{2m}$; $(m \neq 0)$

$h = \dfrac{6m-3}{4m^2}$, $k = \dfrac{6m+3}{4m}$, Now eliminating m and

we get

$\Rightarrow \; 3h = 2(-2k^2 + 9k - 9)$

$\Rightarrow \; 4y^2 - 18y + 3x + 18 = 0$

Q4 (31)

$21\alpha + 12(2-\alpha) + 3 = 0$

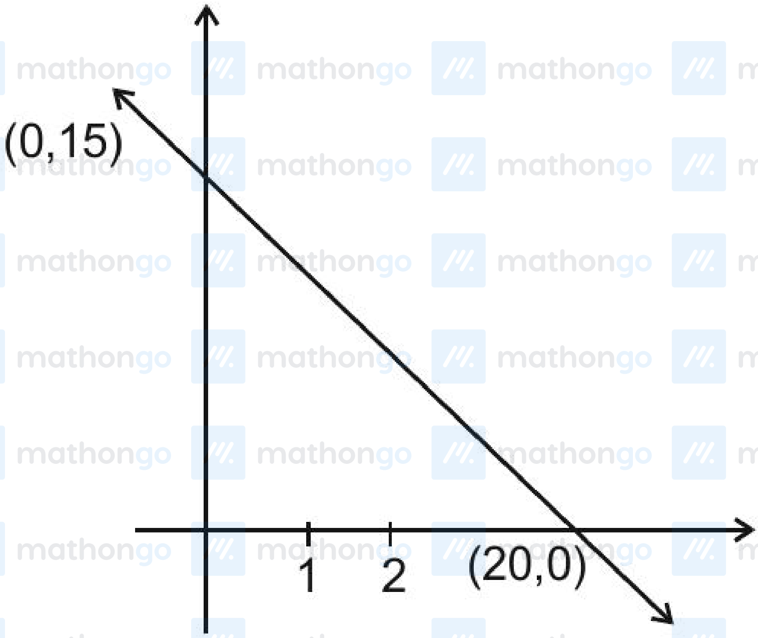
$21\alpha + 24 - 12\alpha + 3 = 0$

$9\alpha + 27 = 0$

$\alpha = -3, \beta = 5$

So $BC = \sqrt{122}$ and $(BC)^2 = 122$

If $x = 1,\ y = \frac{57}{4} = 14.25$



$(1, 1)\ (1, 2) - (1, 14) \quad \Rightarrow 14$ pts.

If $x = 2,\ y = \frac{27}{2} = 13.5$

$(2, 2)\ (2, 4)\ \ldots.\ (2, 12) \quad \Rightarrow 6$ pts.

If $x = 3,\ y = \frac{51}{4} = 12.75$

$(3, 3)\ (3, 6) - (3, 12) \quad \Rightarrow 4$ pts.

If $x = 4, y = 12$

$(4, 4)\ (4, 8) \quad \Rightarrow 2$ pts.

If $x = 5.\ y = \frac{45}{4} = 11.25$

$(5, 5), (5, 10) \quad \Rightarrow 2$ pts.

If $x = 6,\ y = \frac{21}{2} = 10.5$

$(6, 6) \quad \Rightarrow 1$ pt.

$(0, 0) \rightarrow 1$ pt.

If $x = 7,\ y = \dfrac{39}{4} = 9.75$

Q5 (2)

Slope of reflected ray $= \tan 60° = \sqrt{3}$

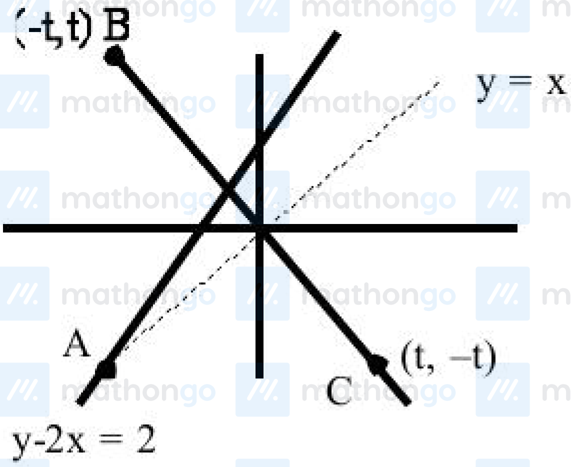
Line $y = \dfrac{x}{\sqrt{3}}$ intersect $y + x = 1$ at $\left(\dfrac{\sqrt{3}}{\sqrt{3}+1}, \dfrac{1}{\sqrt{3}+1}\right)$

Equation of reflected ray is

$$y - \frac{1}{\sqrt{3}+1} = \sqrt{3}\left(x - \frac{\sqrt{3}}{\sqrt{3}+1}\right)$$

Put $y = 0 \Rightarrow x = \dfrac{2}{3+\sqrt{3}}$

Q6 (3)



At A $x = y$

$Y - 2x = 2$

$(-2, -2)$

Height from line $x + y = 0$

$$h = \frac{4}{\sqrt{2}}$$

$$\text{Area of } \Delta = \frac{\sqrt{3}}{4} \frac{h^2}{\sin^2 60} = \frac{8}{\sqrt{3}}$$

Q7 (10)

$S_1 : y^2 = 2x \quad S_2 : x^2 + y^2 = 4x$

$P(2,2)$ is common point on S_1 & S_2

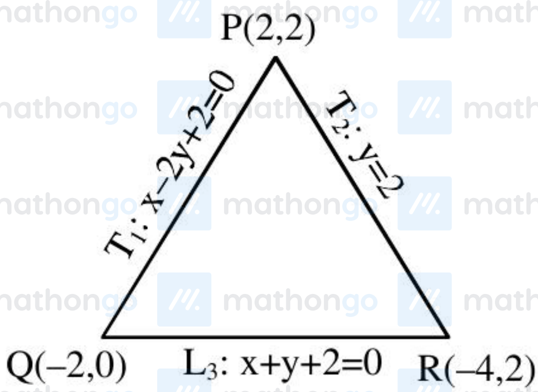
T_1 is tangent to S_1 at P $\quad \Rightarrow T_1 : y.2 = x + 2$

$$\Rightarrow T_1 : x - 2y + 2 = 0$$

T_2 is tangent to S_2 at P $\quad \Rightarrow T_2 : x.2 + y.2 = 2(x+2)$

$$\Rightarrow T_2 : y = 2$$

& $L_3 : x + y + 2 = 0$ is third line



$PQ = a = \sqrt{20}$

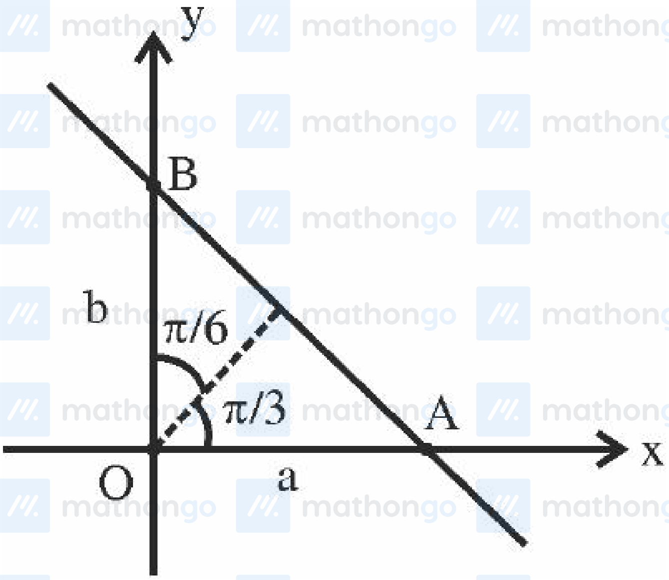
$QR = b = \sqrt{8}$

$RP = c = 6$

Area $(\Delta PQR) = \Delta = \frac{1}{2} \times 6 \times 2 = 6$

$\therefore\ r = \frac{abc}{4\Delta} = \frac{\sqrt{160}}{4} = \sqrt{10} \Rightarrow r^2 = 10$

Q8 (1)



Equation of straight line : $\frac{x}{a}+\frac{y}{b}=1$

Or $x\cos\frac{\pi}{3}+y\sin\frac{\pi}{3}=p$

$$\frac{x}{2}+\frac{y\sqrt{3}}{2}=p$$

$$\frac{x}{3p}+\frac{y}{2p}=1$$

Comparing both : $a=2p, b=\frac{2p}{\sqrt{3}}$

Now area of $\Delta OAB=\frac{1}{2}\cdot ab=\frac{98}{3}\cdot\sqrt{3}$

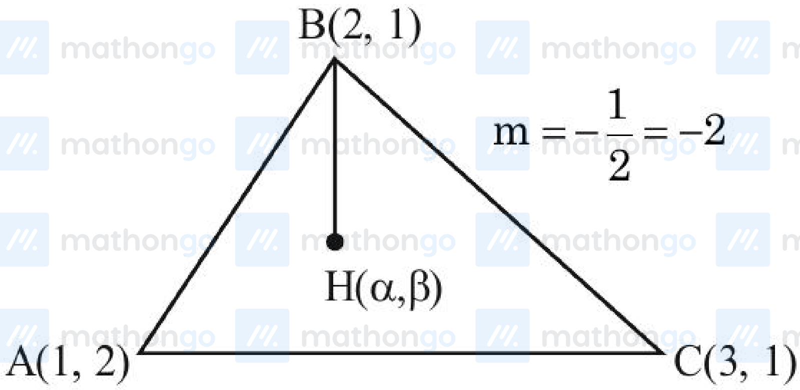
$$\frac{1}{2}\cdot 2p\cdot\frac{2p}{\sqrt{3}}=\frac{98}{3}\cdot\sqrt{3}$$

$$p^2=49$$

$$a^2-b^2=4p^2-\frac{4p^2}{3}=\frac{2}{3}4p^2$$

$$= \frac{8}{3} \cdot 49 = \frac{392}{3}$$

Q9 (4)



$$m = -\frac{1}{2}$$

Here $mBH \times mAC = -1$

$$\left(\frac{\beta-3}{\alpha-2}\right)\left(\frac{1}{-2}\right) = -1$$

$\beta - 3 = 2\alpha - 4$

$\beta = 2\alpha - 1$

$m_{AH} \times m_{BC} = -1$

$\Rightarrow \quad \left(\frac{\beta-2}{\alpha-1}\right)(-2) = -1$

$\Rightarrow \quad 2\beta - 4 = \alpha - 1$

$\Rightarrow \quad 2(2\alpha - 1) = \alpha + 3$

$\Rightarrow \quad 3\alpha = 5$

$$\alpha = \frac{5}{3}, \beta = \frac{7}{3} \Rightarrow H\left(\frac{5}{3}, \frac{7}{3}\right)$$

$$\alpha + 4\beta = \frac{5}{3} + \frac{28}{3} = \frac{33}{3} = 11$$

$$\beta + 4\alpha = \frac{7}{3} + \frac{20}{3} = \frac{27}{3} = 9$$

$x^2 - 20x + 99 = 0$

Q10 (4)

Equation of the pair of angle bisector for the homogenous equation $ax^2 + 2hxy + by^2 = 0$ is given as

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

Here $a = 2$, $h = ½$ & $b = -3$
Equation will become

$$\frac{x^2 - y^2}{2 - (-3)} = \frac{xy}{1/2}$$

$x^2 - y^2 = 10xy$

$x^2 - y^2 - 10xy = 0$